Geography of Ethiopia and The Horn





Course name: Geography of Ethiopia and The Horn
Credit Hours: 3
Semester: I
Course code: (GeES1101)
Academic Year: 2025/26
Target group: Freshman student

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Introduction to the Course

- **Until 2005,** this course were given to all freshman students of the country.
- It was interrupted in relation to **curriculum revision.**
- After a recent curriculum revision (2011 EC).
- One of the common course **- Geography of Ethiopia and the Horn**
- The course is important to familiarize students with **geographic concepts mainly in relation to Ethiopia** and the **Horn of Africa**

Con't…….

The course mainly consists of four major parts (Objectives)

- Basic skills of **geographic position**
- **Physical background** and **resource endowment** of Ethiopia
- **Demographic characteristics**
- Various **economic activities**

Course Content

In order to achieve the above stated objectives, the course has eight chapters.

Chapter One: Introduction

1.1. Geography: Definition, Scope and Themes

1.2. Location, Shape and Size of Ethiopia and the Horn

1.3. Basic Skills of Map Reading

Chapter Two: The Geology Of Ethiopia And The Horn

2.1. The Geologic Processes

2.2. The Geological Time scale and Age Dating Techniques

2.3. Resulting Landforms of Ethiopia and the Horn

2.4. Rock and Mineral Resources of Ethiopia

Con't…….

Chapter Three: Topography of Ethiopia and the Horn

3.1. The Physiographic Divisions of Ethiopia

3.2. The Impacts of Relief on Biophysical and Socioeconomic Conditions

Chapter Four: Drainage Systems and Water Resource of Ethiopia and the Horn

4.1. Major Drainage System of Ethiopia

4.2. Water Resources: Rivers, Lakes and Sub-Surface Water

4.3. Water Resources Potentials and Development in Ethiopia

Con't…….

Chapter Five: The Climate of Ethiopia and the Horn

5.1. Elements and Controls of Weather and Climate

5.2. Spatiotemporal Patterns and Distribution of Temperature and Rainfall in Ethiopia

5.3. Climate Change/Global Warming: Causes, Consequences and Response Mechanisms

Chapter Six: Soils, Natural Vegetation and Wildlife Resources of Ethiopia and the Horn

6.1. Ethiopian Soils

6.2. Natural Vegetation of Ethiopia

6.3. Wild Life/Wild Animals in Ethiopia

Con't…….

Chapter Seven: Population of Ethiopia and the Horn

7.2. Population Data: Uses and Sources

7.3. Population Dynamics: Fertility, Mortality and Migration

7.4. Age and Sex Structure of Ethiopian Population

7.5. Population Distribution in Ethiopia

7.6. Socio-cultural Aspects of Ethiopian Population

Chapter Eight: Economic Activities in Ethiopia

8.1. Primary Economic Activities

8.2. Secondary Economic Activities

8.3. Tertiary Economic Activities

Chapter One: Introduction

Chapter Objectives

At the end of this chapter, you should be able to:

- Comprehend the meaning, scope and approach of Geography.
- Explain the themes of Geography.
- Explicate the implications of location, shape and size of Ethiopia and the Horn on the physical environment, socioeconomic and political aspects.
- Acquire basic skills of map reading.

1.1. Geography: Definition, Scope, Approach and Themes

1.1.1. Meaning of Geography

- No universally accepted definition: Reasons
  - ✓ **the dynamic nature of the discipline** and
  - ✓ **the changes in its scope and method of study.**

? Do you remember the definition given by different scholars

- Working definition.
- Geography is the **scientific study of the Earth** that describes and analyses **spatial** and **temporal** variations of **physical** and **human** phenomena, and their **interrelationships** and **dynamism** over the surface of the Earth.

1.1.2. The Scope, Approaches and Themes of Geography

Scope

- The scope of Geography is **the surface of the Earth**.

❖ The interface of the **four major subsystems**;

- ✓ land, water, living things and air commonly known as **geo-spheres**.

1. **Atmosphere (troposphere)**: all the airs in the earths system
2. **Lithosphere**: the solid part of the earth's surface. Eg. Rocks and related materials
3. **Hydrosphere**: Water bodies of the earth
4. **Biosphere**: includes all the planet living things

Con't…….

Approach:

- Two major approaches; Topical-regional

Topical (Systematic) Approach: applies a specific geographical element or phenomenon over a defined geographical unit.

- **Eg**. The Geography of hunger, the Geography of Climate

Regional Approach: studies the main characteristics of each region of the world

- **Eg.** The geography of Ethiopia, The Geography of Africa

Two major approaches; Topical-regional



Topical Approach

- **Description:** This approach focuses on specific environmental themes or issues, analyzing them independently of geographic boundaries.
- **Example Topics:**
  - **Climate Change:** global temperature trends or carbon emissions.
  - **Biodiversity:** Examining the loss of species in diverse ecosystems worldwide.
  - **Water Management:** Investigating water scarcity solutions, such as desalination or rainwater harvesting, for application in different regions.

Regional Approach

- **Description:** This approach examines environmental issues within a specific geographical area, considering the unique context of that location.
- **Example Regions:**
  - **South America (Amazon Basin):** Research on deforestation rates and conservation strategies.
  - **North Africa:** Investigating the effects of desertification and water scarcity.
  - **East Asia:** Analyzing air pollution sources and policies

Themes of Geography: Five basic themes.

1. Location: is defined as a particular place or position.

-Absolute Location: The exact position of a place, often given in latitude and longitude (40.7128° N, 74.0060° W).

-Relative Location: Describes a place in relation to other places (e.g., New York City is located northeast of Philadelphia).

2. Place: Describes the physical and human characteristics that make a location unique.

Eg. of Characteristics:

-Physical: Landforms, climate, vegetation, wildlife (e.g., the Sahara Desert is known for its arid climate and sand dunes).

-Human: Language, religion, architecture, population, and cultural practices (e.g., Paris is known for the Eiffel Tower, its art scene, and French culture).

Con't…….

3. Human-Environment Interaction:

- ceaseless interaction
- No other species that has lived on our planet has a profound effect on the environment as humans.
- The interaction involves three distinct aspects,
  a. **Dependency**: dependent on nature for a living.
  b. **Adaptation**: relates to how humans **modify themselves**, their **lifestyles** and their **behavior** to live in a new environment with new challenges.
  c. **Modification**: allowed humans to "**conquer**" (overcome challenges) the world for their comfortable living.

Con't…….

4. Movement: entails to the translocation of **human beings**, their **goods**, and **their ideas** from one end of the planet to another.

- **Movement has also three Aspects**
  - ✓ physical movement of **people**
  - ✓ the transport of **goods**
  - ✓ the flow of **ideas**

5. Region: is a geographic area having distinctive characteristics that distinguish itself from adjacent unit(s) of space.

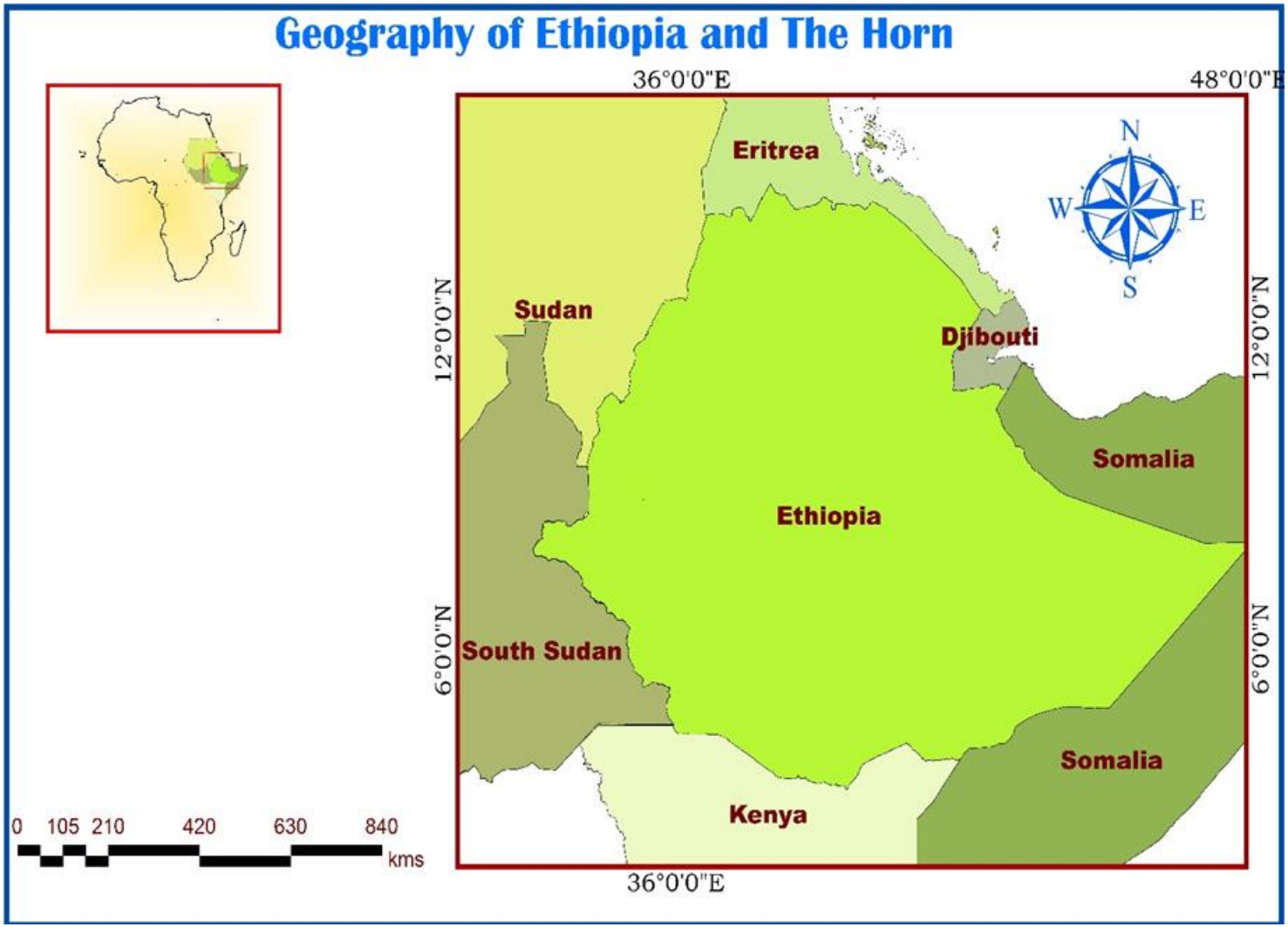
- The region can be two forms;
  - ✓ It could be **a formal region** that is characterized by homogeneity in terms of a certain phenomenon.
  - ✓ It can also be **a functional or nodal region** characterized by functional interrelationships.

1.2. Location, Shape and Size of Ethiopia and the Horn

The Horn of Africa

- a region of **eastern Africa**
- It is the **easternmost extension** of African landmass
- Home to the countries of **Djibouti, Eritrea, Ethiopia**, and **Somalia**,
- Although the countries of the region **share many common features**, there is also **great diversity** among them, rendering **each country unique** in many respects.
- In terms of size, **Ethiopia is the largest** of all the Horn of African countries, while **Djibouti is the smallest**.

Con't……


Geography of Ethiopia and The Horn
36°0'0"E
48°0'0"E
12°0'0"N
6°0'0"N
Eritrea
Sudan
Djibouti
Somalia
Ethiopia
South Sudan
Somalia
Kenya
N
W
E
S
0 105 210 420 630 840
kms

Con't…….

1.2.1. Location of Ethiopia

- Location is **defined as the position of a place or an object on the earth's surface.**
- It is the **identification of a place**, a country or a region in the world.
- All places have **two different types** of location.

1. Absolute (Astronomical) and
2. Relative locations

1. Absolute location: also called **Astronomical** or **mathematical** location

- refers to the **exact position of a place** or **an object**.
- using the lines of **latitudes** and **longitudes**.

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- Astronomically, Ethiopia is a landlocked country located between

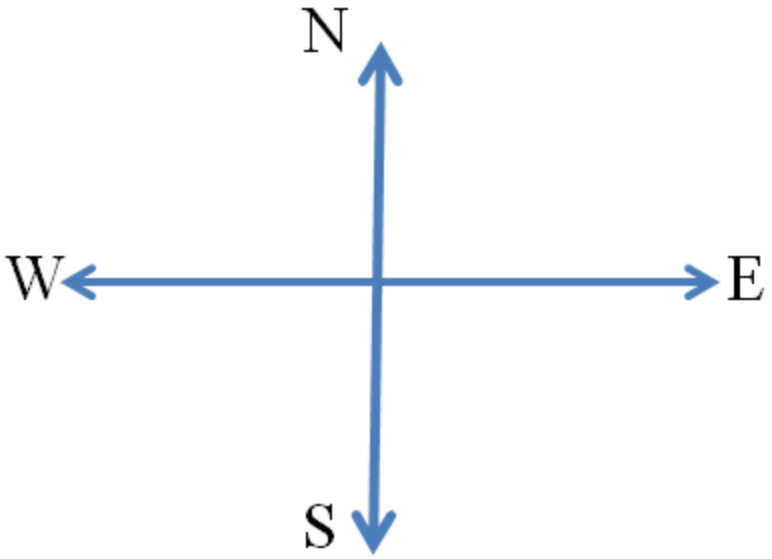
  3°N to **15°N** latitudes and **33°E** to **48°E** longitudes

  ✓ Moyale is the southernmost tip

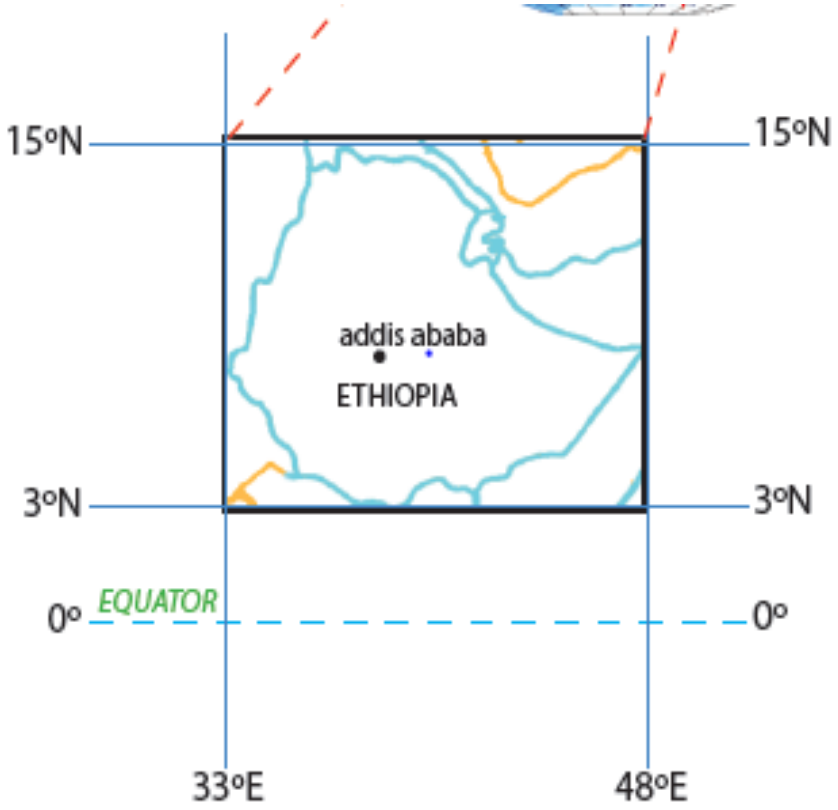
  ✓ Bademe is the northernmost tip

  ✓ Akobo westernmost tip

  ✓ Ogaden the easternmost tip



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East - West distance (15^0)

North-South distance (12^0)

Scale 1:15,000,000

Source: *Adapted from School Atlas of Ethiopia*

Con't…….

- The latitudinal and longitudinal extensions are important in two ways.
  - ✓ **First**, as a result of its latitudinal extension the country **experiences tropical climate** and
  - ✓ **Second**, due to its longitudinal extension there is a difference of **1 hr' between the most easterly** and **most westerly points** of the country.
- It is only for convenience that the **3 hours-time zone** is used in all parts of the country.

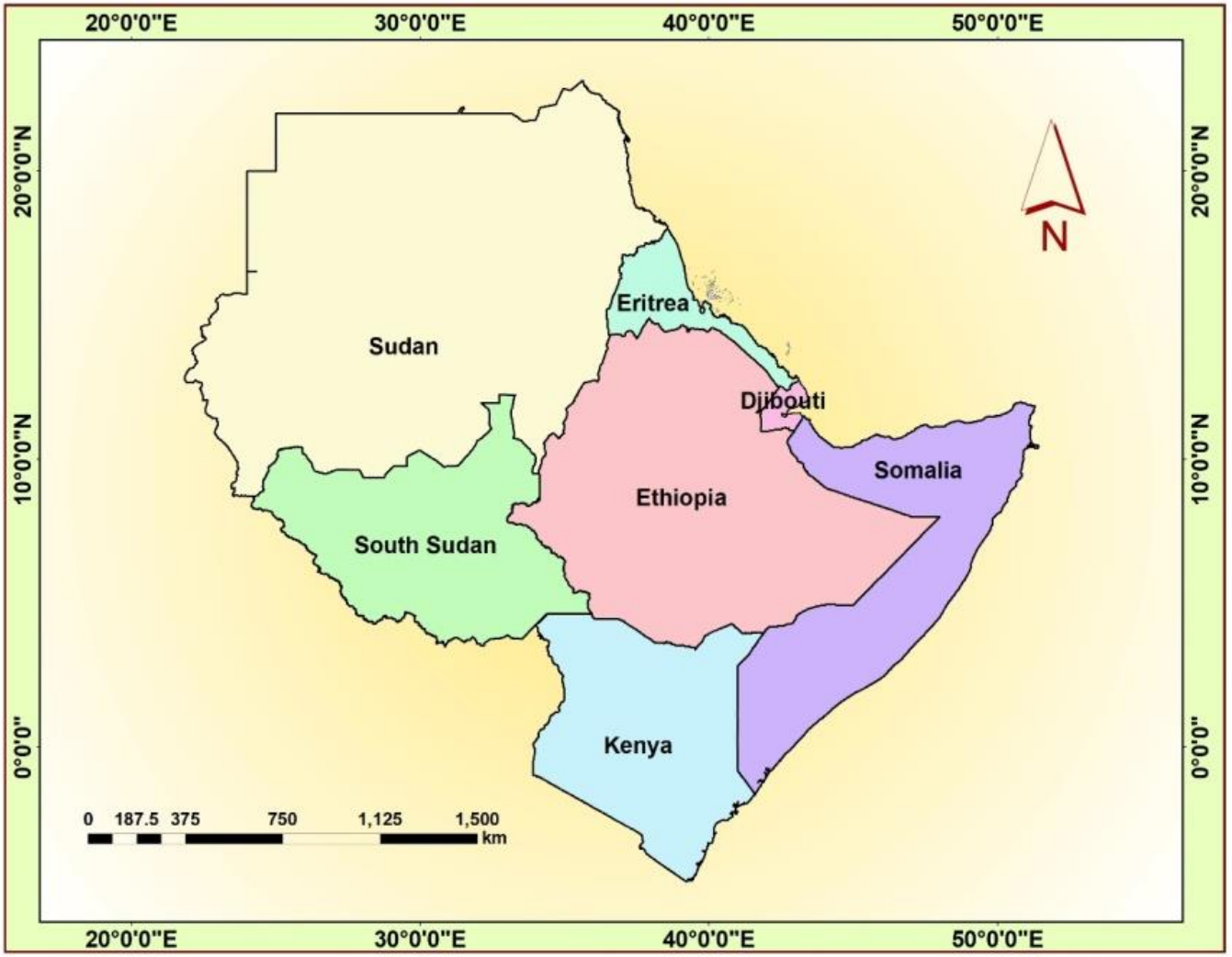
2. Relative location: expresses the location of countries or places with **reference to the location of other countries** (vicinal), **landmasses** or **water bodies**.

Con't…….

Table 1.1: The relative location of Ethiopia

Vicinal location	In relation to water bodies & land masses
➲ Sudan to the west and northwest ➲ South Sudan to southwest ➲ Djibouti to the east ➲ Somalia to the east and southeast ➲ Eritrea to the north and northeast ➲ Kenya to the south	➲ In the Horn of Africa ➲ Southwest of the Arabian Peninsula ➲ South of Europe ➲ Northwest of the Indian Ocean ➲ In the Nile Basin

Con’t…….


20°0'0"E
30°0'0"E
40°0'0"E
50°0'0"E
20°0'0"N
10°0'0"N
0°0'0"
N
Eritrea
Sudan
Djibouti
Somalia
Ethiopia
South Sudan
Kenya
0 187.5 375 750 1,125 1,500
km

The location of Ethiopia has three major implications: **Climate, Socio-cultural** and **Political**

1. Climate



Altitude



- **Diverse Climate Zones**:

-Ethiopia's geographical features, including mountains, plateaus, and valleys, create a variety of climate zones, from tropical in the lowlands to temperate in the highlands. This diversity affects agriculture, water resources, and biodiversity.

- **Rainfall Patterns**:

-Ethiopia relies heavily on seasonal rains, particularly the Kiremt (June to September) and Belg (February to May) seasons. Changes in these patterns due to climate change can lead to droughts or floods, affecting food security.

- **Ecosystem Resilience**:

-The variety of ecosystems supports a range of flora and fauna but also makes them vulnerable to climate variability and environmental degradation.

Con't…….

2. Socio-cultural – recipient of the earliest religion

Orthodox



Islam



Buddhism



Cultural Diversity:

- Ethiopia is home to over 80 ethnic groups, each with its own language, traditions, and customs.
- This diversity contributes to rich cultural heritage but also presents challenges in terms of national unity and integration.

Religious Significance:

- Ethiopia has a unique religious landscape, being one of the oldest Christian nations and also home to significant Muslim populations.
- This religious diversity influences social interactions and political affiliations.

Historical Legacy:

- Ethiopia's long history, including its status as one of the few African nations to resist colonization, contributes to a strong national identity and pride among its people

Con't…….

3. Political: center of politics



Is a strategic location to control Europe, Africa and Asia

Nearness to the Red Sea

- **Geopolitical Significance**:

-Ethiopia's location in the Horn of Africa makes it strategically important, influencing regional politics and security dynamics.

-It shares borders with several countries and plays a key role in regional organizations.

- **Conflict and Tension**:

-Ethnic diversity can lead to tensions and conflicts, particularly over resource allocation and political representation.

-Historical grievances and power struggles often manifest in political disputes.

- **Development Challenges**:

-Political stability is essential for development, but issues such as governance, human rights, and economic inequality can hinder progress.

-Ethiopia’s political landscape is shaped by the need to balance ethnic representation and national unity

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4. Land-lockdeness:

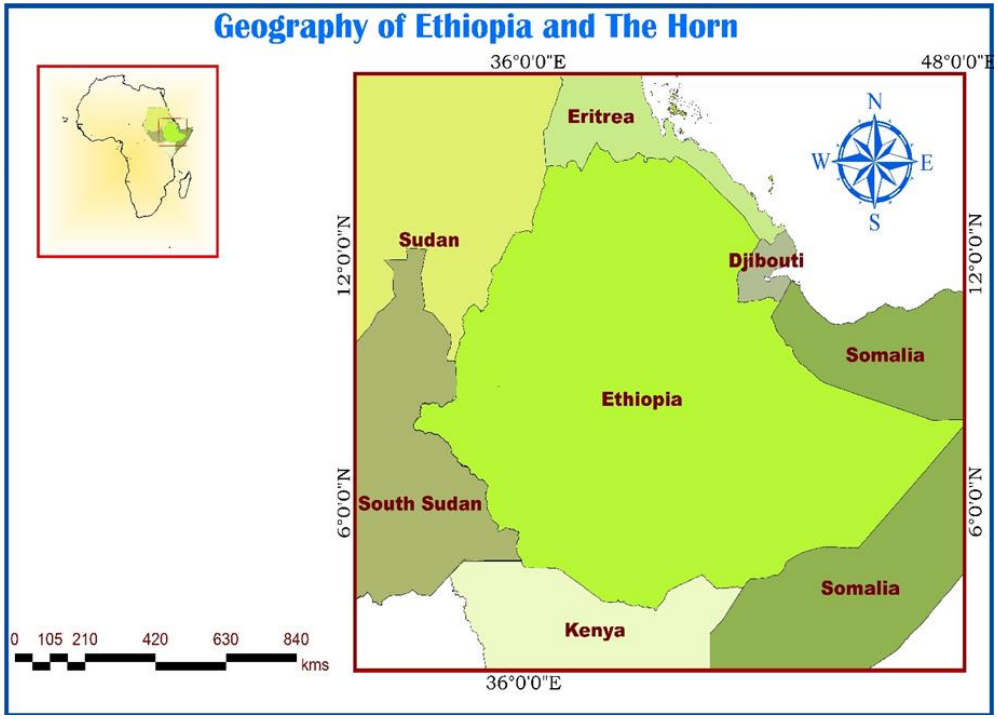


The Sea Ports of Djibouti

Con't…….

1.2.2. Areal Size of Ethiopia

- **1,106,000** KM^2
- **8th largest** in Africa
- **25th** in the World.
- **1,639 kilometers** East-West
- **1,577 kilometers** North-South.
- About **0.7% water bodies**.



Con't…….

Table 1.2: Advantages and disadvantages of Ethiopia's large

Advantages	Disadvantages
Possess diverse agro ecological zones	Demands greater capital to construct infrastructural facilities
Variety of natural resources	Requires large army to protect its territory
Own extensive arable land	Difficult for effective administration
Have larger population size	Difficult for socio-economic integration
Home for diverse cultures	
Greater depth in defense external invasion	

Con't…….

1.2.3. The shape of Ethiopia and its Implication

Countries of the World have different kinds of shape that can be divided into five main categories:

- ✓ compact,
- ✓ fragmented,
- ✓ elongated,
- ✓ perforated, and
- ✓ protruded.

- These shapes have implications on **defense**, **administration** and **economic integration** within a country.

Con't…….

1. Compact shape countries: The distance from the **geographic center** of the state to any of the borders **does not vary greatly.**



Poland



Ethiopia

Con't…….





2. Fragmented shape countries: They are divided from their other parts by either **water**, **land** or **other countries**.

3. Elongated shape countries: They are **geographically long** and **relatively narrow** like Chile.

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5. Perforated Shape Countries: A country that completely surrounds another country like the republic of South Africa



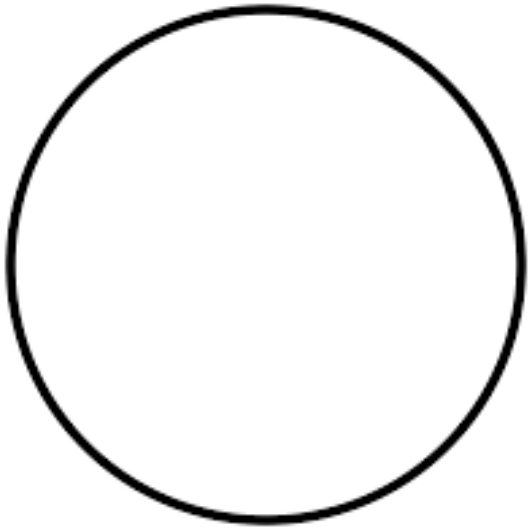
Con't…….

6. Protrude shape countries: Countries that have one portion that is much more elongated than the rest of the country like Myanmar and Eritrea.





Con't…….



How can we Measure the Shape of a country?

- There are **various ways** of measuring shape of countries.
- These measures are known as the **indices of compactness**.
- These indices measure the **deviation of the shape** of a country from a circular shape, which is the most compact shape.

Con't…….

- There are **four most commonly used measures** of compactness. These are:

1. **Area-Boundary ratio:** the ratio of area of country to its boundary length:.
   - The higher the A/B ratio, the greater the degree of compactness.

Example: The total area of Ethiopia is 1,106,000 km^2 with a boundary length of 5260 km.

- Hence **Areas – boundary ratio** is calculated as the total area of the country divided by the boundary length:

  1,106,000 km^2 /5260 km = **210.27 km**

Con't…….

2. **Boundary-Circumference ratio:** the ratio of boundary length of a country to the circumference of a circle having the same area as the country itself.

- It measures how far the boundary of a country approximates the circumference of a circle of its own size.
- Therefore, the nearer the ratio to 1 the more compact the country is.

Example: The boundary length of Ethiopia is 5260 km. Hence in order to calculate the circumference of a circle having the same area with Ethiopia:

Con't…….

Solution:

- Circumference is calculated as 2IIr, where r is radius,
- Radius is calculated as the square root of Area divided by II. Therefore:
- Circumference of a circle = 2IIr
- Radius, $r = \sqrt{\frac{\text{Area}}{\pi}} = \sqrt{\frac{1{,}106{,}000\text{km}^2}{3.14}} = 593$ kms
- Boundary length of Ethiopia = 5260kms
- Hence, B/C ratio of Ethiopia $\frac{B}{C} = \frac{5260 \text{ kms}}{2 \times 3.14 \times 593\text{kms}} = \frac{5260}{2 \times 3.14 \times 593} = \underline{1.41}$

Con't…….

3. **Area-Circumference ratio**: the ratio of the area of the country to the circumference of the smallest inscribing circle.

- It compares the area of the country with the circumference of a circle that passes touching the extreme points on the boundary of the country.
- To calculate the A/C ratio for Ethiopia, we'll follow these steps:
  - **Calculate the radius of the inscribing circle** using the area of the country.
  - **Calculate the circumference of the inscribing circle**.
  - **Calculate the A/C ratio**.

Calculations

1. **Calculate the Radius r:**

$$r = \sqrt{\frac{1,106,000}{\pi}} \approx \sqrt{\frac{1,106,000}{3.14159}} \approx 593.34\,\text{km}$$

2. **Calculate the Circumference C:**

$$C = 2\pi r \approx 2 \times 3.14159 \times 593.34 \approx 3,728.06\,\text{km}$$

3. **Calculate the Area-Circumference Ratio:**

$$\text{A/C Ratio} = \frac{1,106,000}{3,728.06} \approx 296.67$$

The higher the A/C ratio, the greater the degree of compactness

4. **Area-Area (A/A') ratio: t**he ratio of the actual area of a country to the smallest possible inscribing circle.

- The area of the inscribing circle is the area of the smallest possible circle whose circumference passes through the extreme points on the boundary.
- Half-length of the longest distance between two extreme points gives radius of the inscribing circle.

How to calculate?

- **Area of Ethiopia (A)**: 1,106,000 km2
- **Area of the smallest inscribing circle (A')**: This is calculated based on the radius of the circle that can enclose Ethiopia.

Calculations

Let's perform the calculations:

1. **Calculate the Area of the Inscribing Circle (A')**: Using the previously calculated radius:

$$A' = \pi(593.34)^2$$

2. **Calculate the A/A' Ratio**:

$$\text{A/A' Ratio} = \frac{1,106,000}{A'}$$

1. **Calculate the Radius r**:

$$r = \sqrt{\frac{1,106,000}{\pi}} \approx \sqrt{\frac{1,106,000}{3.14159}} \approx 593.34\,\text{km}$$

2. **Calculate the Area of the Inscribing Circle (A')**:

$$A' = \pi r^2 \approx \pi \times (593.34)^2 \approx 1,107,474.59\,\text{km}^2$$

3. **Calculate the A/A' Ratio**:

$$\text{A/A' Ratio} = \frac{1,106,000}{1,107,474.59} \approx 0.9987$$

The nearer the ratio to 1, the more compact the country is.

The shape…

Table 1.3: Ethiopia's shape compared to its neighbors in the Horn.

Country	Area(km^2)	Boundary (km)	A/B ratio	B/C ratio	A/C ratio
Ethiopia	1,106,000	5,260	210.27	1.41	296.61
Djibouti	22,000	820	26.83	1.56	41.83
Eritrea	117,400	2,420	48.51	1.99	96.83
Kenya	582,644	3,600	161.85	1.33	215.28
Somalia	637,657	5,100	125.03	1.80	225.22

Source: Bekure (1982, 1983)

1.3. Basic Skills of Map Reading

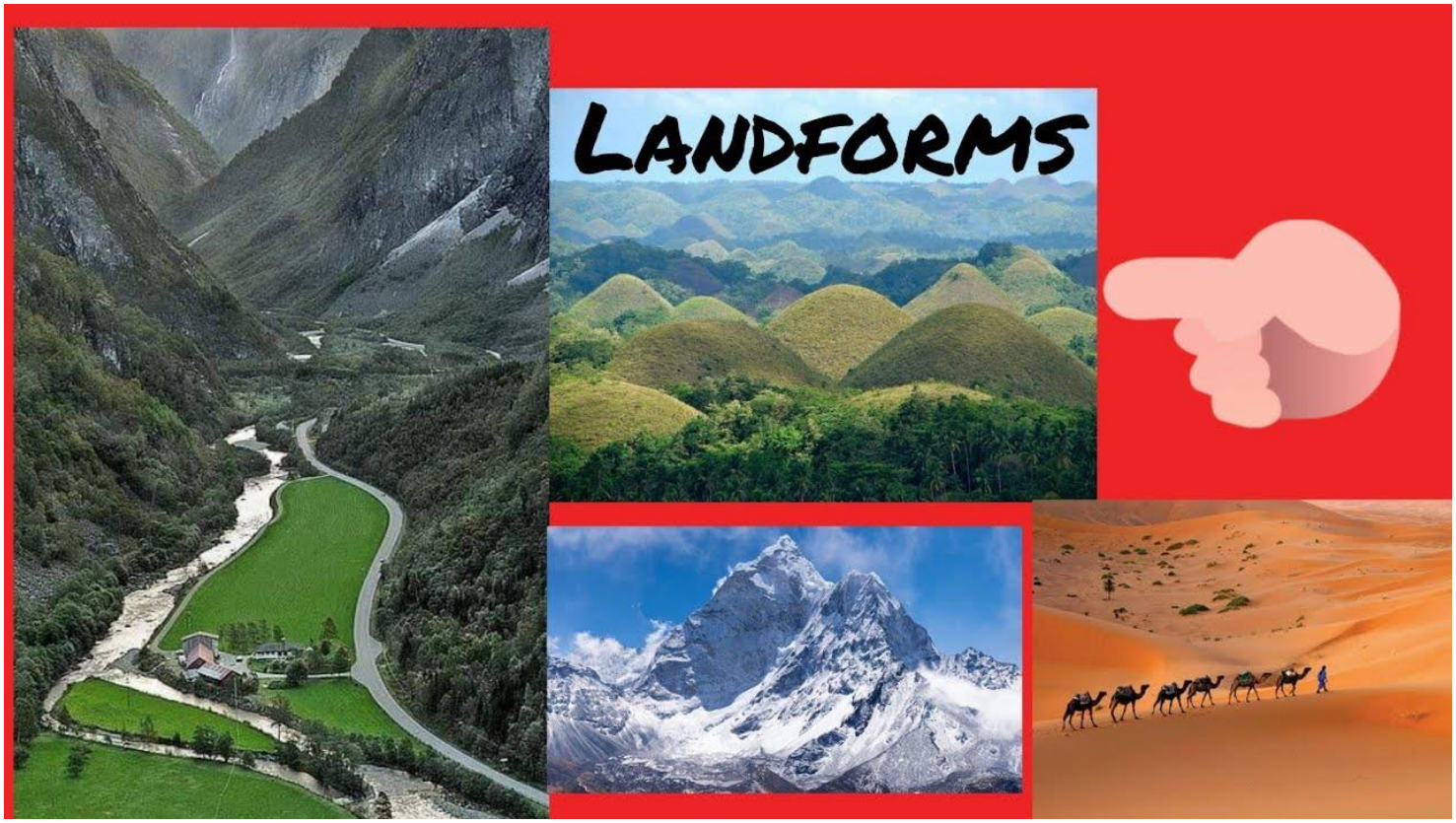
What is a Map?

- A map is **a two-dimensional** scaled representation of **part or whole of the Earth surface** on a **flat body** such as piece of paper, black board, wood or cloth.



1.3. Basic Skills of Map Reading

- Map reading encompasses a systematic identification of **natural features** and **manmade** features.
- **Natural features** include mountains, plateaus, hills, valleys, river, ocean, rocks, plain etc.



- **manmade features** include roads, railway, buildings, dam etc.









- Since these features cannot easily be observed and interpreted in real landscapes, maps are essential to Geographers.

Con't……..

Importance of maps

- Provide the basis for **making geographical details** of regions
- Maps are **powerful tools** for making spatial analysis of geographical facts of areas represented.
- Maps are useful for giving **location of geographical features**
- Map makes storage of the geographical data of areas represented.
- Maps are potentially used to asses' **reliable measurements of the geographical features**.

  The measurements can be of area size, distance etc.

Con't……..

Types of Map

- There are many types of maps according to **their purpose** and **functions**.
- For the purpose of this course, **topographical** and **statistical** maps are considered.

1. Topographical maps: Topographic maps depict one or more **natural** and **cultural** features of an area.

- They could be **small, medium** or **large** scale depending on the size of the area represented.
- Contents of topographical maps depend on **purpose of a map**, **scale of a map**, **date of compilation**, and **nature of the land** represented.

2. Special purpose/statistical maps: These are maps, which show distribution of different aspects such as **temperature, rainfall, settlement, vegetation** etc.

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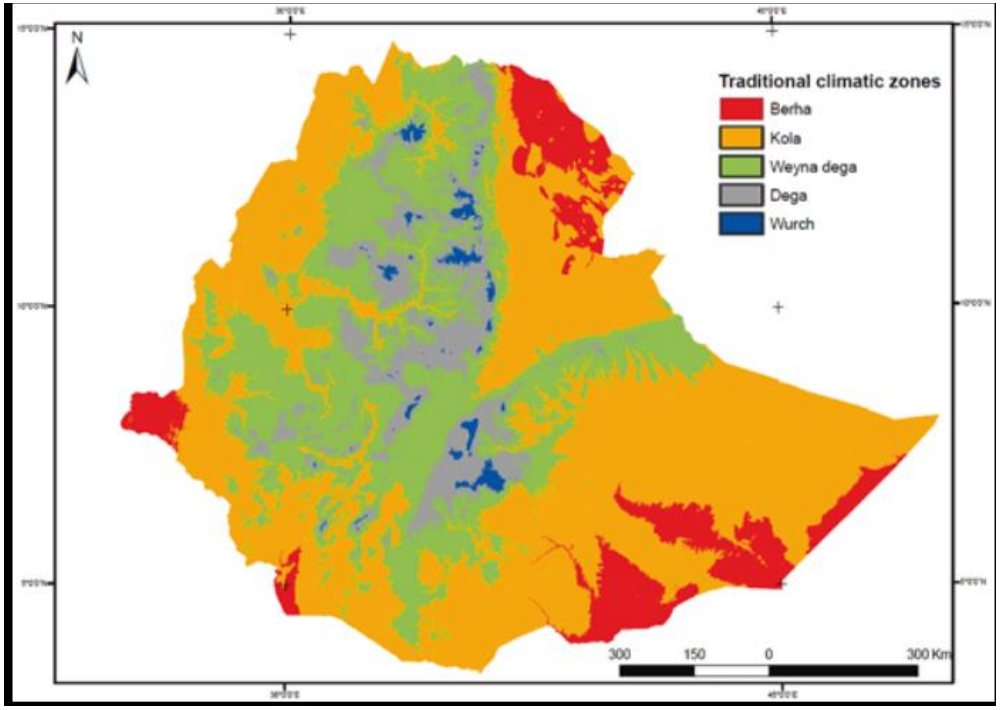
Marginal Information on Maps (Elements of Maps)

- Marginal information is shown on a map to enable the reading and interpretation of the geographical information of an area represented. This includes:
- **Title:** It is the heading of the given map which tells what the map is all about.

  Ex: Map of Ethiopia

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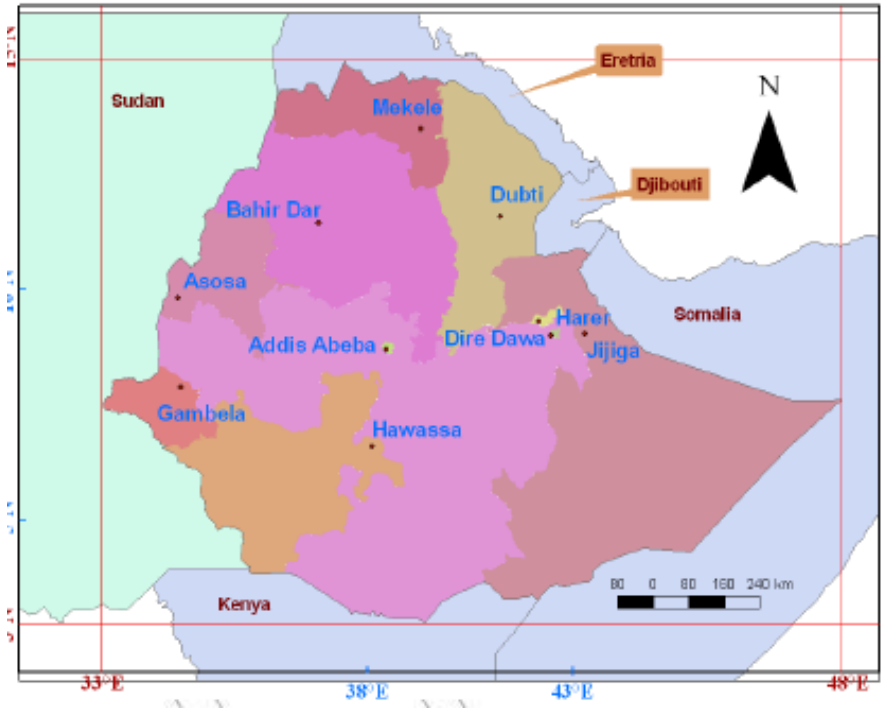
- **Key (legend):** It is the list of all convectional symbols and signs shown on the map with their interpretation.



- **Margin:** Is the frame of the map. It is important for showing the end of the mapped area.

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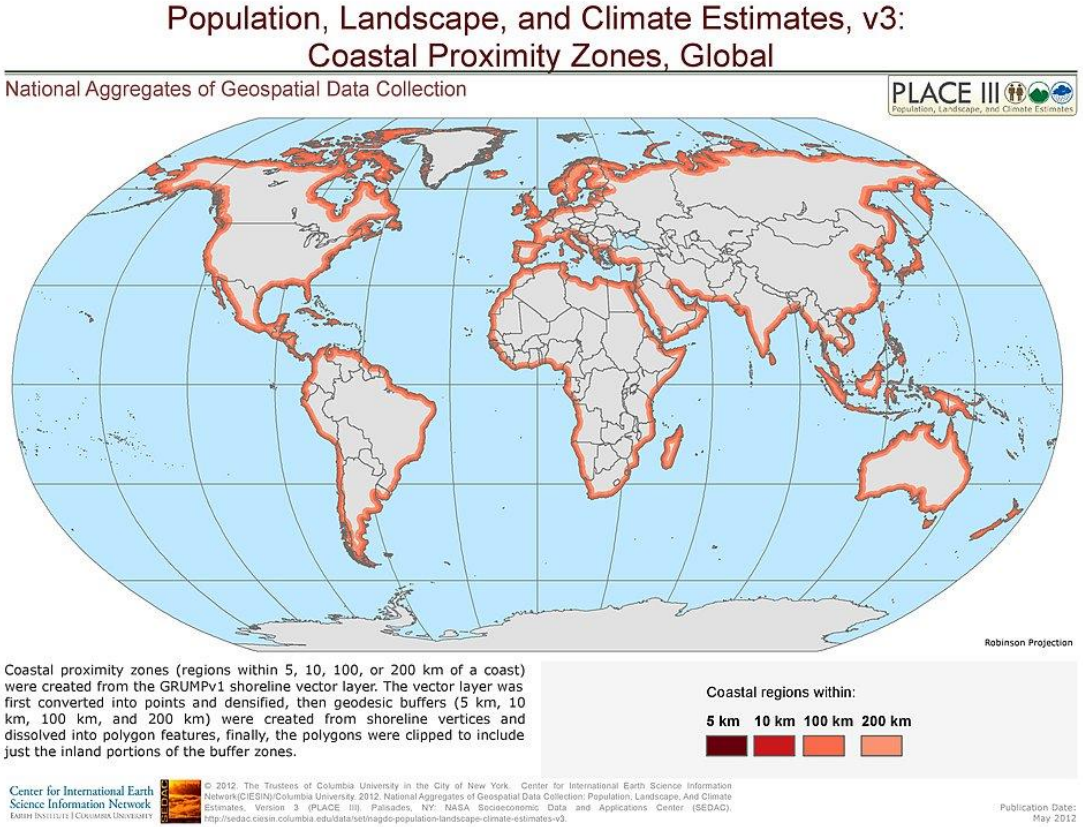
- **Scale:** It is the ratio between the distance on the map and the actual ground distance.



- **North arrow:** It is indicated with the north direction on a map; used to know the other important directions of the mapped area like east, south, and west.

Con't……..

- **Date of compilation:** It is a date of map publication. This enables map users to realize whether the map is updated or outdated.



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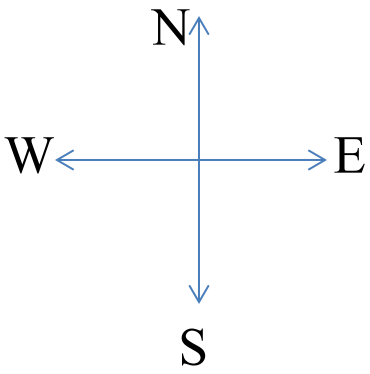
Con't……..

Basic Principles of Map Reading

- In developing map reading abilities and skills, certain basic principles must be applied by the map-reader to translate map symbols into landscape images.
  - ✓ Map Readers must have ideas about the **symbol** and also **the real World** (landscapes).

✝ ☾

  - ✓ Secondly, knowledge of directions is an important principle in reading maps.



Summary of the Lesson

Main points of the lesson

1.1. Geography: definition, scope, approach and theme

1.2. Geographic position of Ethiopia and the Horn

Location of Ethiopia and the horn

Absolute and Relative location

implications of Ethiopia's location

Areal size of Ethiopia

- ✓ Advantages and Disadvantages of Ethiopia's areal size

Shape of Ethiopia

- ✓ the five categories of shape
- ✓ measurement of country's shape

1.3. Basic Skills of Map Reading

Definition, importance, type, elements and principles